SmartHire — On-Campus Recruitment and Preparation System

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ABSTRACT

- **Background:** College campus placement drives require intense coordination between student applicants, company recruiters, and the Training & Placement Office (TPO). Traditional hiring drives rely on Google Forms, emails, and spreadsheet trackers, leading to manual sorting delays, lack of candidate feedback, and disorganized records across academic departments.
- **The Problem:** Manual workflows lack automated test proctoring, require third-party paid external links for coding evaluations, and provide students with zero pre-application feedback on how well their resume matches the recruiter's specific job description (JD).
- **SmartHire Solution:** An end-to-end web and mobile-assisted on-campus recruitment and preparation platform built with React.js, Node.js, Express, and Supabase PostgreSQL. It unifies job posting, student eligibility screening, online coding tests via JDoodle, browser webcam proctoring, instant ATS resume scoring, and mobile phone push notifications into a single integrated system.
- **Project Objective & Scope:** To replace manual campus placement coordination with an automated, cheat-resistant recruitment and preparation ecosystem featuring verified student KYC, cloud sandboxed coding exams, and direct mobile push notifications.

INTRODUCTION

- **Three Role-Based Portals:** SmartHire provides three separate interfaces: *Student Portal & App* (job applications, push alerts on phone, coding tests, ATS score check), *Recruiter Portal* (post jobs, review applicants, advance stages), and *College Admin Portal* (verify students, approve drives, track placement statistics).
- **Online Test Proctoring:** Includes assessments with webcam face detection (using `face-api.js`), tab-switch warnings, and fullscreen enforcement to ensure fair evaluation without demanding heavy media streaming servers.
- **Integrated Code Runner:** Connects to the JDoodle API to compile and execute student programs in C++, Java, Python, and JavaScript against automated test cases with runtime and memory metrics.

Student Web & App	Recruiter Portal	T&P Admin Portal
Phone Push Alerts • Tests • ATS Resume Checker	Post Drives • Review Pool • Stage Shortlist	Student KYC • Job Approval • Placement Stats

LITERATURE SURVEY

Feature / Aspect	Traditional System	Commercial Portals	SmartHire (Ours)
Placement Tracking	Google Sheets & Forms	Generic Job Boards	**Dedicated 3-Role Unified Portal**
Coding Assessments	Third-party external links	Paywalled separate tools	**Built-in JDoodle Multi-Lang Runner**
Exam Proctoring	Manual in-person invigilation	Heavy video streaming	**Browser Face Detection & Tab Blur Check**
Resume Screening	Manual resume reading	Opaque AI Auto-Rejection	**Instant Keyword Match & Skill Feedback**
Drive Notifications	Notice boards & WhatsApp	Delayed email digests	**Mobile Phone Alerts & Email Notifications**
Application Status	Offline / Verbal updates	Static status text	**Live 5-Stage Tracking (Applied → Selected)**

METHODOLOGY / IMPLEMENTATION

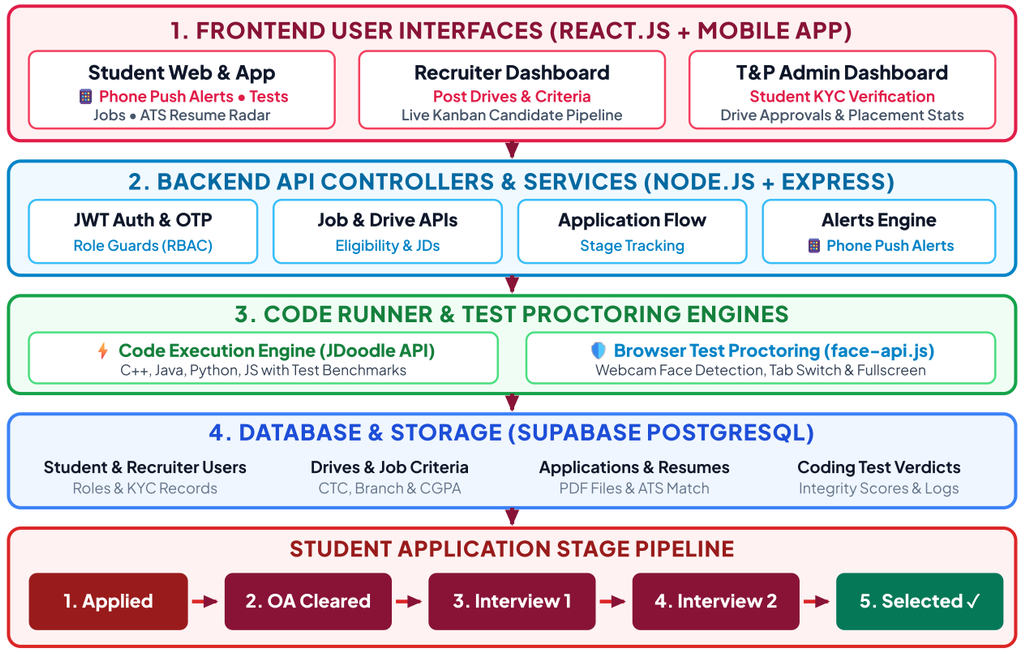


Fig 1: SmartHire Multi-Tier System Architecture & Application Pipeline Workflow

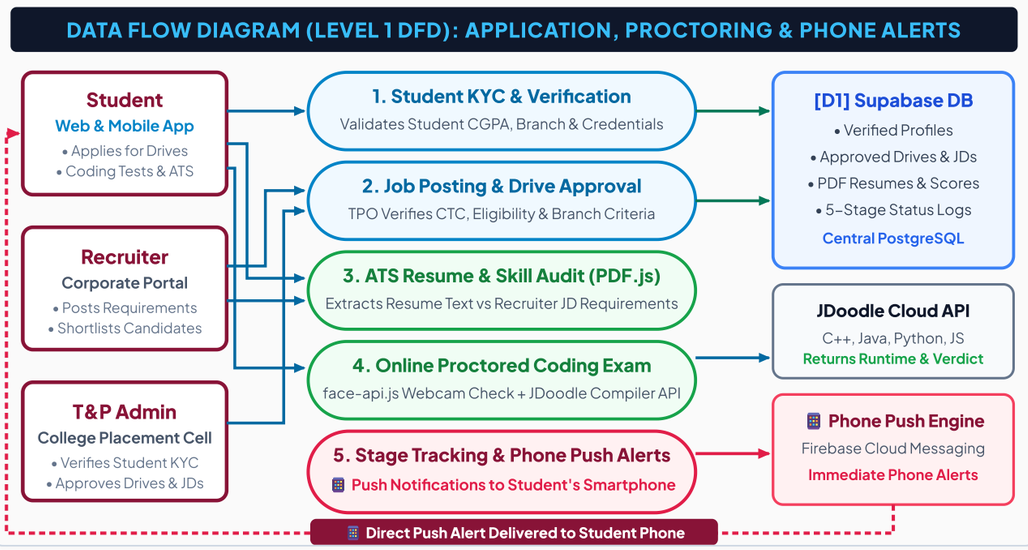


Fig 2: SmartHire Data Flow Diagram (DFD)

Client-Side AI Proctoring (face-api.js)
Real-time webcam face presence detection, multiple face alerts, window blur and tab-switching violation tracking.

Multi-Lang Code Sandbox (JDoodle API)
Automated compilation and execution for C++, Java, Python, and JavaScript against test suites with runtime metrics.

Browser ATS Resume Audit (PDF.js)
Client-side text parsing compares candidate skills directly with recruiter job requirements, returning match percentage.

5-Stage Pipeline & Phone Alerts
Recruiters track candidates from Applied to Selected; automated push notifications instantly alert students on their smartphones.

RESULTS & ANALYSIS

RESULT: PROCTORED CODING ARENA — 100% INTEGRITY

TwoSum.cpp • C++ (JDoodle API) — ● Camera Active (face-api.js)

```
vector<int> twoSum(vector<int>& nums, int target) {
  unordered_map<int,int> map; // O(n) hash lookup
  for(int i=0; i<nums.size(); i++) { ... }
```

✓ All 12/12 Test Cases Passed — Execution Time: 0.04s

RESULT: STUDENT SKILLS RADAR — SCORE: 92/100



DSA & Logic	94%
System Design	88%
Aptitude & Verbal	91%
ATS Skill Match	92/100

RESULT: RECRUITER DRIVE PIPELINE — LIVE STAGE SYNC

Drive: Software Engineer 2026 — 42 Candidates

OA CLEARED	INTERVIEW	OFFERED
18 Students	6 Shortlisted	2 Selected ✓

RESULT: ATS RESUME SKILL AUDIT — MATCH: 88%

Role: Full-Stack Developer — Status: High Match

✓ Matched: React.js, Node.js, Express, PostgreSQL, REST APIs
⚠ Recommended to Add: Docker, TypeScript, Git Actions
Result: Student notified of keyword gaps before drive submission

4	100%	<1s	3
LANGUAGES (C++, JAVA, PY, JS)	BROWSER PROCTORING	ATS SCORE CHECK	DEDICATED PORTALS

CONCLUSION & FUTURE SCOPE

- **Conclusion:** SmartHire successfully replaces scattered spreadsheets and manual placement coordination with a unified, modern on-campus recruitment and preparation system. It bridges students, recruiters, and the college placement cell through verified profiles, transparent application stages, built-in coding tests, and proctoring.
- **Future Scope 1 (Student Mobile App & Push Notifications):** Developing a dedicated cross-platform mobile application so students receive real-time push notifications on their phones for new drive postings, exam schedules, and round-by-round shortlist results.
- **Future Scope 2 (AI Voice Mock Interviews):** Integrating an LLM-assisted conversational voice mock interview practice module to help students prepare for behavioral and technical interviews.
- **Future Scope 3 (NIRF & NAAC Analytics):** Providing automated placement analytics reporting, statistical salary distributions, and instant compliance data exports.

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